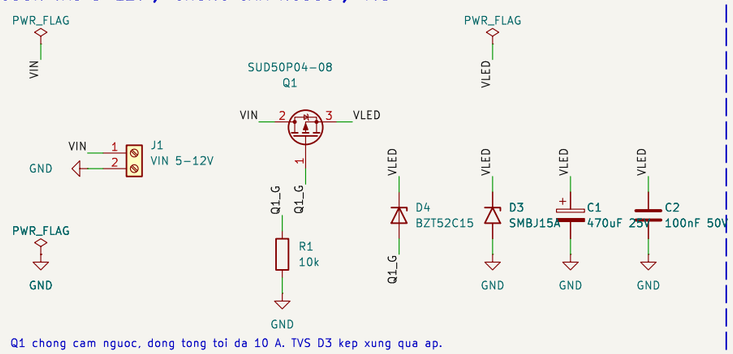
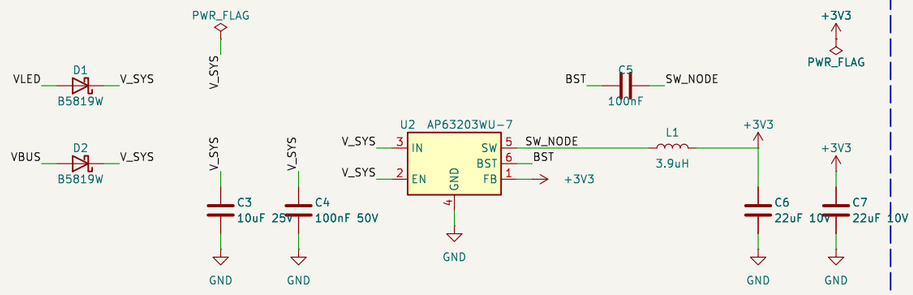
NGUON VAO 5–12V / CHONG CAM NGUOC / TVS

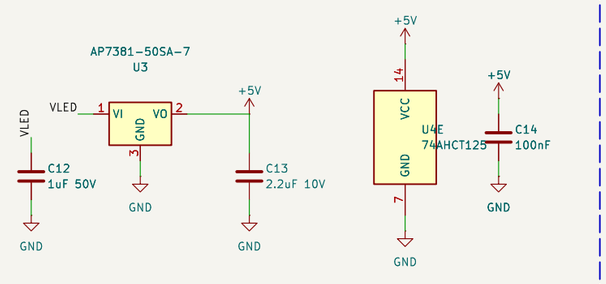


Q1 chong cam nguoc, dong tong toi da 10 A. TVS D3 kep xung qua ap.

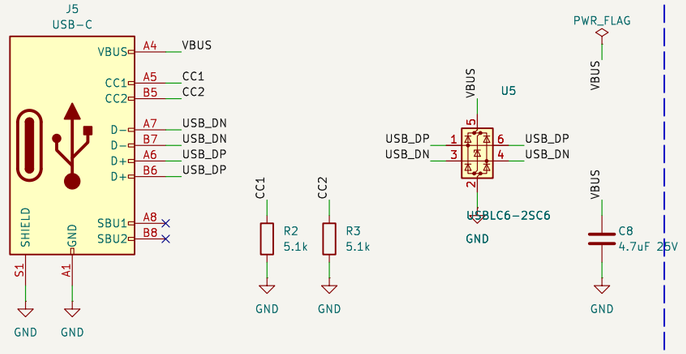
NGUON 3.3V (buck AP63203)



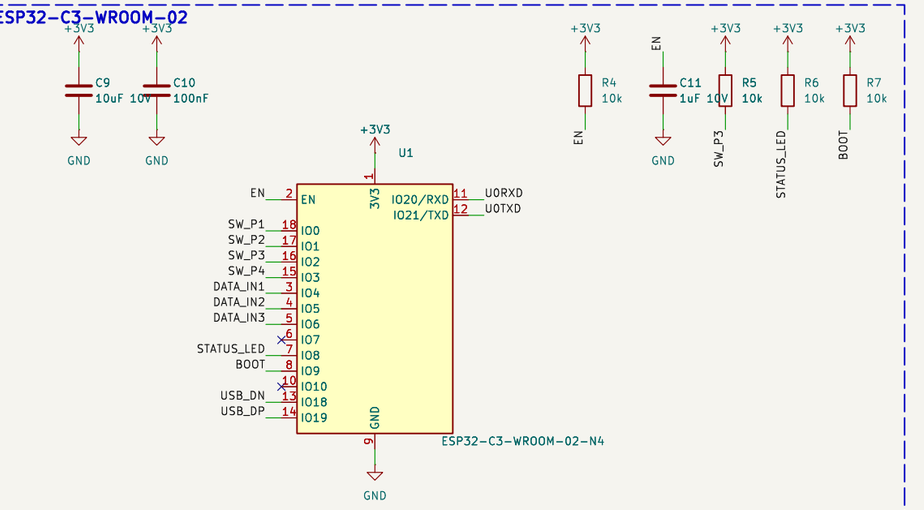
5V CHO IC DICH MUC (chi co khi co nguon LED)



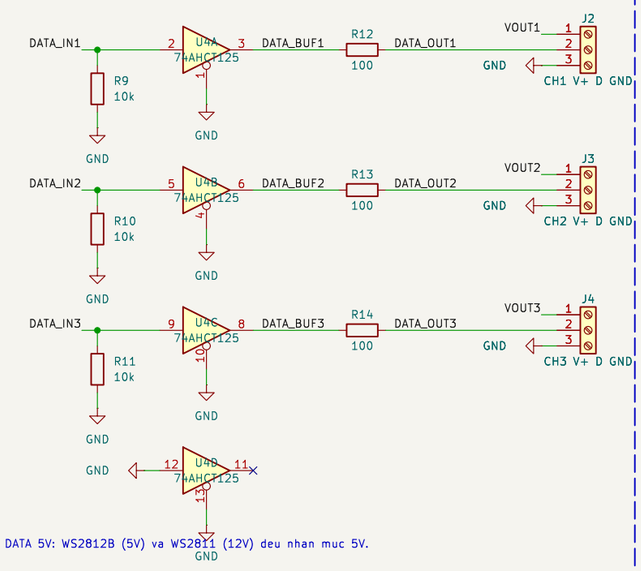
USB-C: NAP CODE / DEBUG (USB Serial/JTAG)



ESP32-C3-WROOM-02

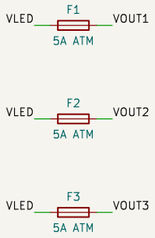


DICH MUC 3.3V -> 5V + 3 KENH WS2812



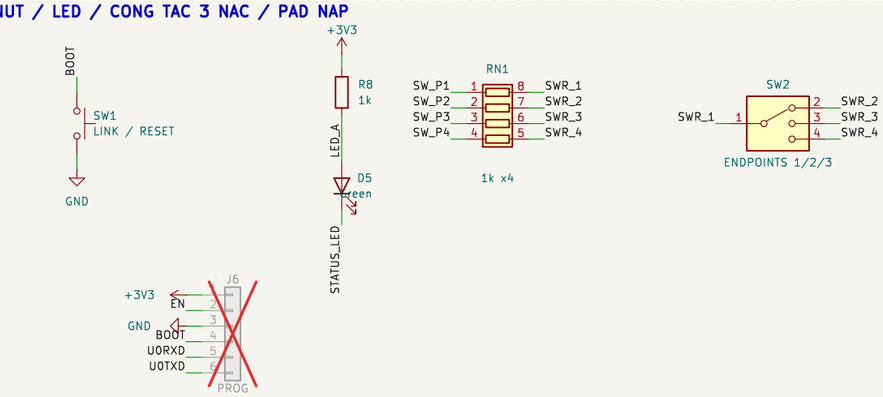
DATA 5V: WS2812B (5V) va WS2811 (12V) deu nhan muc 5V.

CAU CHI TUNG KENH



Moi kenh mot cau chi o to mini (ATM) rieng:
chap 1 kenh khong lam mat cac kenh con lai.

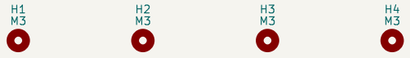
NUT / LED / CONG TAC 3 NAC / PAD NAP



SW1 (GPIO9 = BOOT): bam nhanh = bat/tat hoac vao che do ket noi; giu >5 s = factory reset.
Giu SW1 trong luc cap nguon = vao che do nap ROM.

SW2 -> IO0..IO3 (IO2 la chan strap: R5 keo len, cong tac khong noi GND nen luon o muc 1 khi khoi dong).
Firmware quet ca 4 chan (lan luot keo 1 chan xuong 0, doc 3 chan con lai, bat pull-up noi).
Nac = so thu tu chan lon nhat trong cap dang noi – 1 -> 1, 2 hoac 3 endpoint Matter.
Khong phu thuoc chan chung cua tung loai cong tac SP3T 4 chan.

LO BAT VIT M3



Generated by tools/build_sch.py from tools/design.py
Input 5–12V, 3x WS2812/WS2811 outputs, 1 button, 3-position endpoint switch

Haxeko

Sheet: /
File: c3_ws2812_3ch.kicad_sch

Title: ESP32-C3 WS2812 3-channel Matter LED controller

Size: A3	Date: 2026-10-30	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1